

REFERENCE-FREE SINGING PITCH CORRECTION VIA MUSIC-CONSTRAINED SEQUENCE EDITING

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ABSTRACT

Singing pitch correction is an important task for improving the quality of amateur singing performances, while existing approaches typically rely on external references, such as target melodies, accompaniment tracks, or predefined pitch constraints. However, these references are often unavailable in many open application scenarios, making reference-free singing pitch correction a challenging problem. In this work, we formulate reference-free singing pitch correction as a sequence editing task under music-theoretic constraints, where correction requires determining whether pitch deviations should be modified and how they should be corrected. To address this problem, we propose a framework that integrates singing-context representation learning, hierarchical edit prediction, and music-theoretic constraints. Specifically, we employ a pretrained symbolic music encoder with singing-domain adaptation to model musical context, and jointly model note-level error detection and pitch offset prediction through hierarchical prediction. Furthermore, tonal information is incorporated as a music-theoretic constraint to guide candidate correction without requiring external reference melodies. Experiments on real-world paired singing recordings demonstrate that the proposed method improves note-level pitch accuracy from 73.25% to 83.43%, outperforming a context-based baseline by 3.99 percentage points and achieving a better balance between error correction and preservation of correctly performed notes.

Index Terms— Singing pitch correction, reference-free correction, sequence editing, music representation.

1. INTRODUCTION

Singing voice is one of the most important forms of musical expression, while pitch inaccuracies in amateur performances can significantly degrade perceived quality. Automatic singing pitch correction has become an important component in music production, online singing platforms, and AI-assisted music creation. However, many practical scenarios, including user-generated a cappella recordings, spontaneous humming, melody sketching, and legacy recordings without project files, provide only the vocal performance itself without an explicit reference melody. Without an explicit reference melody, reference-free correction must infer intended pitches from the input’s musical context while preserving expressiveness.

Existing singing pitch correction methods typically rely on explicit musical references or predefined pitch constraints. Commercial tools such as Auto-Tune and Melodyne provide scale- or note-constrained pitch editing[1, 2]. Academic systems likewise exploit auxiliary musical information: Deep Autotuner predicts note-wise pitch shifts from the vocal and accompaniment spectrograms[3], KaraTuner conditions pitch prediction on musical notes[4], and

Neural Singing Voice Beautifier aligns an amateur performance with a professional template pitch curve[5]. More recently, generative models based on conditional variational autoencoders and diffusion have explored natural vocal modification under template-, score-, or MIDI-derived pitch conditions [5, 6, 7]. However, these methods require target melodies, accompaniment tracks, or other external references that are unavailable in many isolated vocal recording scenarios. Recent work has therefore begun to investigate reference-free singing pitch correction, in which plausible pitches are inferred from the intrinsic musical context of the input performance[8].

Reference-free singing pitch correction is fundamentally different from conventional pitch estimation. While pitch estimation focuses on recovering the current pitch from an input signal, reference-free correction additionally requires determining whether a pitch deviation should be modified. Similar edit decision formulations have been widely explored in sequence correction tasks, where error detection and editing operations are explicitly modeled for fine-grained sequence modification [9, 10, 11]. However, singing pitch correction introduces unique musical characteristics that differ from general sequence correction: the validity of a note depends not only on local pitch variation but also on melodic context, tonal structure, and expressive singing characteristics. Music representation models underscore the importance of contextual modeling for music understanding[12], while singing research reveals interactions among pitch, technique, and expressive style [13]. Accordingly, reference-free correction is formulated as a music-constrained sequence editing task informed by music-specific knowledge.

To address these challenges, we propose a reference-free singing pitch correction framework that formulates pitch correction as a music-constrained sequence editing task. Specifically, we adopt MusicBERT, a pretrained symbolic music representation model [14], as the backbone encoder and introduce lightweight singing-domain adapters to obtain contextual representations tailored to singing MIDI. Based on these representations, we design a hierarchical editing strategy that decomposes pitch correction into error detection and pitch offset prediction, explicitly modeling note-level editing decisions. Furthermore, we estimate a tonal prior from the input singing performance itself and use it to rerank candidate corrections, introducing musical structural constraints without requiring target melodies or ground-truth key annotations. Audio examples are available at <https://github.xx.xx.io>.

The main contributions of this work are as follows:

- We propose a reference-free pitch correction framework for real amateur singing that exploits the intrinsic musical context of the input without target melodies or external references.
- We formulate correction as a music-constrained sequence editing task and introduce a hierarchical strategy for note-level error detection and pitch-offset prediction.
- We introduce input-dependent tonal-aware candidate rerank-

ing, which incorporates musical constraints without ground-truth key annotations.

2. METHODOLOGY

2.1. Problem formulation

Given a dry singing vocal recording, reference-free singing pitch correction aims to improve pitch accuracy without requiring target melodies, accompaniment tracks, or other external references. Unlike pitch estimation, correction requires identifying inaccurate deviations and modifying them toward musically appropriate pitches while preserving valid performances.

As illustrated in Fig. 1, we perform correction in a symbolic singing representation space. We use the open-source SOME model to extract note-level MIDI and F_0 contours from singing audio. After symbolic correction, the modified pitches are mapped back to the original F_0 contour, followed by NSF-HiFi-GAN reconstruction. Stable regions are shifted directly, while note boundaries are smoothly interpolated.

Specifically, the extracted singing MIDI sequence is represented as

$$\mathcal{X} = \{x_i\}_{i=1}^N, \quad (1)$$

where N denotes the number of detected notes. Each note is represented as

$$x_i = (p_i, d_i, s_i, e_i), \quad (2)$$

where p_i , d_i , s_i , and e_i denote the pitch, duration, onset position, and ending position of the i -th note, respectively. These symbolic attributes provide musical context for correction, while only the pitch attribute is modified and the temporal structure is preserved.

The objective is to generate a corrected singing representation

$$\mathcal{Y} = \{y_i\}_{i=1}^N, \quad (3)$$

where

$$y_i = (\hat{p}_i, d_i, s_i, e_i), \quad (4)$$

and $\hat{p}_i$ denotes the corrected pitch. The correction process can therefore be formulated as estimating a pitch editing operation

$$\Delta_i = \hat{p}_i - p_i, \quad (5)$$

where Δ_i represents the pitch modification applied to the i -th note.

However, reference-free singing pitch correction is not equivalent to directly predicting the target pitch sequence. Some pitch deviations may correspond to intentional expressive variations rather than errors. The correction process consequently requires an editing decision that indicates whether a note should be modified. We introduce an editing variable

$$z_i = \begin{cases} 1, & \text{correction,} \\ 0, & \text{otherwise,} \end{cases} \quad (6)$$

and formulate the final corrected pitch as

$$\hat{p}_i = \begin{cases} p_i + \Delta_i, & z_i = 1, \\ p_i, & z_i = 0. \end{cases} \quad (7)$$

Thus, we formulate reference-free singing pitch correction as a music sequence editing task, where the model learns when and how to modify pitch deviations from the intrinsic musical context of the singing performance.

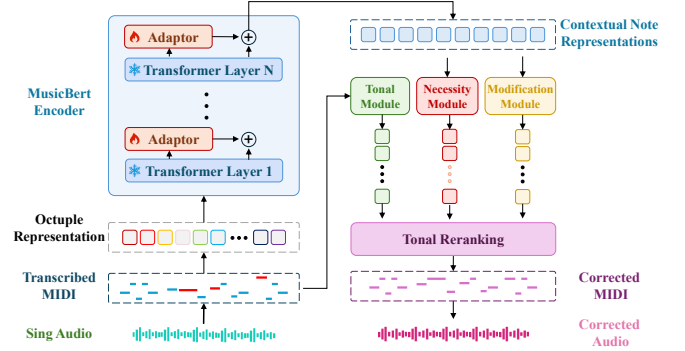


Fig. 1: Overview of the proposed reference-free singing pitch correction framework.

2.2. Singing-aware music context encoder

Reference-free singing pitch correction requires modeling musical context to infer appropriate pitch modifications. We therefore employ a pretrained symbolic music encoder to capture contextual dependencies among musical events. Specifically, we adopt MusicBERT as the backbone encoder[14]. MusicBERT is pretrained on large-scale symbolic MIDI corpora and learns contextual representations of structured musical events through Transformer-based pretraining.

Although MusicBERT provides strong symbolic music representations, it is pretrained on general MIDI data, whereas singing MIDI exhibits different characteristics, including monophonic melodic structures and vocal-specific pitch variations. To bridge this domain gap, we introduce lightweight singing-domain adapters[15] while keeping the pretrained MusicBERT parameters frozen. The adapted representation is

$$\tilde{H} = H + A(H), \quad H = \text{MusicBERT}(\mathcal{X}). \quad (8)$$

One gated adapter is inserted after each of its 12 Transformer blocks:

$$A_l(h) = \sigma(g_l) W_2 \text{Drop}_{0.1}(\text{GELU}(W_1 \text{LN}(h))), \quad (9)$$

where $W_1 \in \mathbb{R}^{64 \times 768}$ and $W_2 \in \mathbb{R}^{768 \times 64}$. The adapted representations are then fed into the subsequent hierarchical edit prediction module.

2.3. Hierarchical correction modeling

Inspired by explicit editing formulations in sequence correction tasks [10, 9], we formulate reference-free singing pitch correction as a hierarchical correction process. Instead of directly predicting the target pitch sequence, the proposed method decomposes correction into two stages: estimating whether a note requires modification and predicting the corresponding pitch modification. Given the contextual representation $\tilde{H} = \{\tilde{h}_i\}_{i=1}^N$, we introduce two lightweight prediction heads. Both heads consist of a feed-forward projection layer followed by softmax/sigmoid activation to predict the corresponding probability distributions. The correction necessity prediction head estimates the probability that a note requires correction:

$$a_i = P(z_i = 1 | \tilde{h}_i) = \sigma(f_{\text{necc}}(\tilde{h}_i)), \quad (10)$$

where $z_i \in \{0, 1\}$ denotes the correction indicator. The pitch modification prediction head predicts a distribution over the nonzero off-

set $\mathcal{D}_{\neq 0} = \{-3, -2, -1, +1, +2, +3\}$, conditioned on correction:

$$b_i(d) = \text{Softmax}\left(f_{\text{mod}}(\tilde{h}_i)\right)_d, \quad d \in \mathcal{D}_{\neq 0}, \quad (11)$$

The two predictions form a factorized pitch-editing distribution:

$$P(\Delta_i = d \mid \tilde{h}_i) = \begin{cases} 1 - a_i, & d = 0, \\ a_i b_i(d), & d \neq 0. \end{cases} \quad (12)$$

The two heads are jointly optimized with classification losses:

$$\mathcal{L} = \mathcal{L}_{\text{nec}} + \mathcal{L}_{\text{mod}}, \quad (13)$$

Specifically, the necessity prediction is trained with binary cross-entropy over all notes, whereas the modification prediction is optimized with cross-entropy only on perturbed notes that require correction. This hierarchical formulation explicitly separates whether a pitch deviation should be modified from how it should be modified, enabling pitch correction while preserving appropriate original notes.

2.4. Tonal-aware reranking

Although the hierarchical correction model predicts pitch modifications from singing context, reference-free correction can still yield multiple plausible candidates without an explicit melody reference. We therefore introduce a tonal-aware candidate reranking strategy. Rather than imposing a hard key constraint, it uses tonal information as a soft prior derived solely from the input singing performance.

We apply the Krumhansl–Schmuckler (KS) key-finding algorithm [16] to a duration-weighted pitch-class histogram u_x . Correlations with the 24 major and minor key templates κ_k are normalized to form a soft key posterior:

$$\rho(k \mid x) = \text{Softmax}_k(\text{corr}(u_x, \kappa_k)/\tau). \quad (14)$$

We set the posterior temperature to $\tau = 0.15$. The posterior induces an input-dependent pitch-class prior

$$r_x(v) = \sum_k \rho(k \mid x) \pi_k(v), \quad (15)$$

Here $v \in \{0, \dots, 11\}$ and $\pi_k(v)$ is a normalized key-specific prior, assigning weights 1.0 to scale pitch classes of key k and 0.15 otherwise. We further compute a song-level confidence $c(x) \in [0, 1]$ from the concentration of $\rho(k \mid x)$ and the number of input notes:

$$c(x) = \left(1 - \frac{\mathcal{H}(\rho)}{\log 24}\right) \min\left(1, \frac{n_x}{2n_{\min}}\right), \quad (16)$$

Here n_x is the input-note count and $n_{\min} = 8$; for fewer notes or a degenerate histogram, we use a uniform posterior and set $c(x) = 0$.

For each note, the editor considers the seven offsets $\mathcal{D} = \{0, \pm 1, \pm 2, \pm 3\}$. We combine the editing distribution with the tonal prior using

$$S_i(d) = \log P(\Delta_i = d \mid \tilde{h}_i) + \lambda c(x) \log r_x((p_i + d) \bmod 12), \quad (17)$$

where $\lambda \geq 0$ controls the global strength of the tonal constraint and is selected on the validation set. The final pitch modification is

$$\hat{d}_i = \arg \max_{d \in \mathcal{D}} S_i(d). \quad (18)$$

This confidence-weighted reranking biases ambiguous candidates toward tonally compatible solutions without forcing pitches to conform to a single detected key.

3. EXPERIMENTS

3.1. Experimental setup

Dataset and Training Protocol. We construct training samples from 16,000 professional singing recordings collected from a private singing corpus, containing approximately 1,060 hours of Chinese and English popular songs. The corpus covers diverse singers and musical styles and is used only for synthetic perturbation generation and model training. The extracted singing MIDI sequences are used to generate synthetic pitch perturbations. Considering the typical vocal range, note pitches are constrained within MIDI 36–84 (C2–C6). For each sequence, 20% of notes are perturbed using six nonzero semitone offsets $\{-3, -2, -1, +1, +2, +3\}$. To simulate different error patterns, 70% of perturbations are independently sampled, while 30% are generated as correlated spans of two to four consecutive notes. Synthetic perturbations are used only for training and controlled validation.

For evaluation, we use a private paired singing dataset containing Chinese and English popular songs. Each song includes amateur and professional performances recorded by vocal students. The paired recordings are aligned using DTW to obtain note-level correspondence. The dataset is split by song, including 20 validation songs (8,299 notes) and 80 held-out test songs (30,682 notes). All hyperparameters, including the tonal reranking strength, are selected using only the validation set. The professional performances are used as evaluation references rather than absolute ground truth, since expressive variations may exist across performances.

Implementation Details. The pretrained MusicBERT encoder is kept frozen during training, while the singing-domain adapters and two prediction heads are optimized. We first train the prediction heads for 12,000 updates, followed by an additional 25,000 updates to jointly optimize the adapters and prediction heads with the frozen MusicBERT backbone. The detection and offset prediction losses are equally weighted. The model checkpoint is selected according to validation performance, and the tonal reranking strength is also determined using the validation set. We use AdamW with cosine schedules, with initial learning rates of 10^{-4} for the task heads and 10^{-5} for the adapters. Each GPU processes 32 sequences with a maximum length of 8,192 field tokens.

Baselines and Evaluation Metrics. We compare our method with two reference-free baselines: an independent reimplementation of BERT-APC[8], which predicts note pitches from musical context, and a key-based pitch snapping method that adjusts notes toward the nearest pitch classes according to the estimated key. We report note-level exact pitch accuracy (Acc.) as the primary metric. To evaluate the correction–preservation trade-off, we additionally report Error Repair and Harm Rate, measuring the recovery of incorrect notes and the undesired modification of correct notes, respectively.

3.2. Objective evaluation

Table 1 summarizes the objective comparison on the real paired singing test set. Compared with the original amateur performances, the proposed method improves note-level pitch accuracy from 73.25% to 83.43%, demonstrating the effectiveness of reference-free pitch correction in real singing scenarios. Compared with BERT-APC, which predicts note pitches based on musical context, our method achieves a further improvement of 3.99 percentage points in accuracy. The proposed method also substantially increases the Error Repair rate from 26.18% to 46.81%, indicating that explicitly modeling correction necessity enables more effective identification and correction of inaccurate notes. Although Key

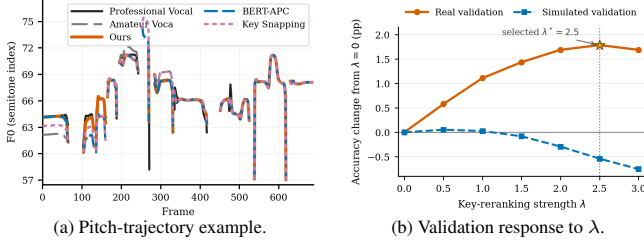


Fig. 2: Qualitative and validation analyses.

Table 1: Objective results on the held-out real test set (%).

Method	Acc. $\uparrow$	Δ Acc. $\uparrow$	Repair $\uparrow$	Harm $\downarrow$
input	73.25	0.00	—	—
Key Snapping	71.76	-1.49	20.76	9.62
BERT-APC	79.44	+6.19	26.18	1.11
Ours	83.43	+10.18	46.81	3.19

Snapping repairs 20.76% of erroneous notes, its 9.62% Harm Rate reduces accuracy to 71.76%, showing the limitation of hard tonal constraints. In contrast, our method combines contextual prediction with tonal-aware reranking, achieving stronger correction capability while maintaining a controlled Harm Rate. These results demonstrate the importance of modeling both correction necessity and modification selection in reference-free singing pitch correction. A representative pitch-trajectory example is shown in Fig. 2(a).

3.3. Ablation study

We conduct ablation studies to investigate the contribution of singing-domain adaptation, hierarchical correction modeling, and tonal-aware reranking. Table 2 summarizes the results on the held-out real test set.

Singing-domain adaptation: We remove the adapter modules and optimize only the prediction heads on top of frozen MusicBERT representations. Removing the adapters reduces note accuracy from 83.43% to 76.87%, showing that adaptation to singing-specific MIDI characteristics is important for reference-free pitch correction.

Hierarchical correction modeling: We compare the proposed factorization with a flat seven-class head that directly predicts $\Delta \in \{0, \pm 1, \pm 2, \pm 3\}$. The flat head attains 82.35% accuracy but repairs fewer erroneous notes than the hierarchical model (39.32% versus 46.81%). The hierarchical formulation increases Harm Rate moderately, reflecting a more active editing policy and the resulting correction–preservation trade-off.

Effect of Tonal-aware Reranking: To evaluate the contribution of tonal information, we remove the tonal-aware reranking module and compare the performance with the full model. As shown in Table 2, removing tonal reranking decreases note-level accuracy from 83.43% to 81.52% and Error Repair from 46.81% to 35.76%. This demonstrates that tonal information provides useful musical priors for selecting correction candidates. We further analyze the influence of tonal-constraint strength λ on synthetic perturbations and real paired singing data, as shown in Fig. 2(b). Tonal constraints provide limited improvement on synthetic perturbations, where randomly generated pitch errors are less correlated with musical structure. In contrast, real singing performances benefit more from tonal information, indicating that realistic singing deviations have stronger relationships with the underlying tonal context. The performance

Table 2: Ablation results on the held-out real test set (%).

Method	Acc. $\uparrow$	Δ Acc. $\uparrow$	Repair $\uparrow$	Harm $\downarrow$
w/o adapters	76.87	+3.62	34.10	7.52
Flat edit head	82.35	+9.10	39.32	1.93
w/o reranking	81.52	+8.27	35.76	1.77
Ours	83.43	+10.18	46.81	3.19

Table 3: Subjective evaluation on a five-point scale. Higher is better.

Method	Pitch Corr. $\uparrow$	Nat. $\uparrow$	Overall $\uparrow$
Input	3.10	3.97	3.25
Key Snapping	3.26	3.80	3.67
BERT-APC	3.46	3.85	3.47
Ours	3.59	3.93	3.94

reaches the best trade-off at $\lambda = 2.5$. A moderate tonal constraint improves candidate selection, whereas overly strong constraints can reduce correction flexibility by forcing predictions toward the estimated tonal structure. This result supports confidence-aware soft tonal constraints rather than hard pitch restrictions.

3.4. Subjective evaluation

To evaluate the perceptual quality of the corrected singing audio, we conduct a subjective listening test with 24 participants with musical backgrounds. Each participant evaluates five randomly selected singing samples generated by different methods using a five-point scale. The evaluation considers three aspects: *Pitch Correctness*, which measures perceived pitch improvement; *Naturalness*, which evaluates the preservation of natural vocal characteristics; and *Overall Quality*, which reflects the overall listening preference.

As shown in Table 3, the proposed method obtains the highest mean scores for all three perceptual aspects. Compared with the input performances, our method improves Pitch Correctness from 3.10 to 3.59 and Overall Quality from 3.25 to 3.94. Meanwhile, its Naturalness score remains close to that of the input recordings (3.93 versus 3.97), indicating that correction does not substantially reduce perceived naturalness. Compared with BERT-APC and Key Snapping, our method provides better perceptual pitch improvement while maintaining higher naturalness. These results suggest that the proposed framework improves singing pitch quality while preserving the expressive characteristics of the original performance.

4. CONCLUSIONS

In this paper, we investigated reference-free singing pitch correction and formulated it as a music-constrained sequence editing task. We proposed a framework that integrates singing-context representation learning, hierarchical correction modeling, and confidence-aware tonal reranking to infer and modify inaccurate pitches from the input performance itself. Experiments on real paired amateur–professional singing recordings demonstrate that the proposed method improves both objective correction performance and perceptual quality, achieving a better balance between error correction and preservation of the original performance. Future work will explore more comprehensive singing representation learning and jointly model pitch, rhythm, and individual singing characteristics, towards automatic singing performance refinement with improved musicality and expressiveness.

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